

CS & IT ENGINEERING



COMPUTER ORGANIZATION AND ARCHITECTURE

CPU & Control Unit

Lecture No.- 3

By- Vishvadeep Gothi sir



Recap of Previous Lecture



Topic

Data Path

Topic

Control Unit

Topic

Hardwired Control Unit

Topics to be Covered



Topic

Microprogrammed Control Unit

Topic

RISC vs CISC

Topic

Byte Ordering



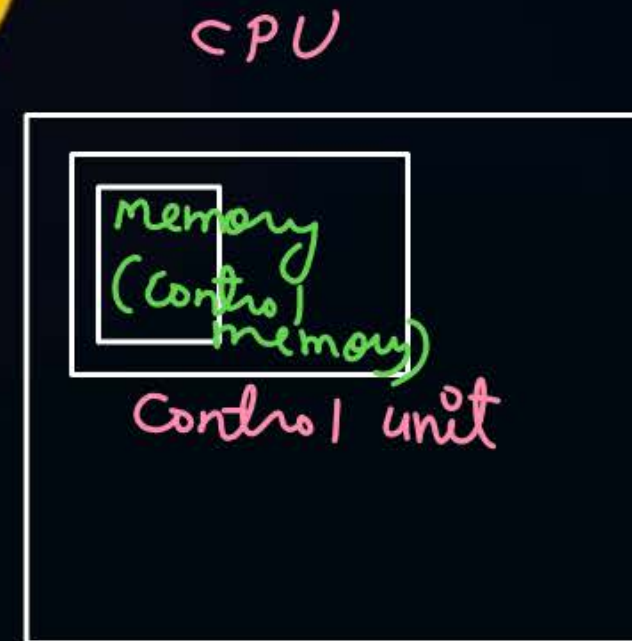
Topic : Micro-Programmed Control Unit

Control logic is implemented with micro-programs.

Advantage:

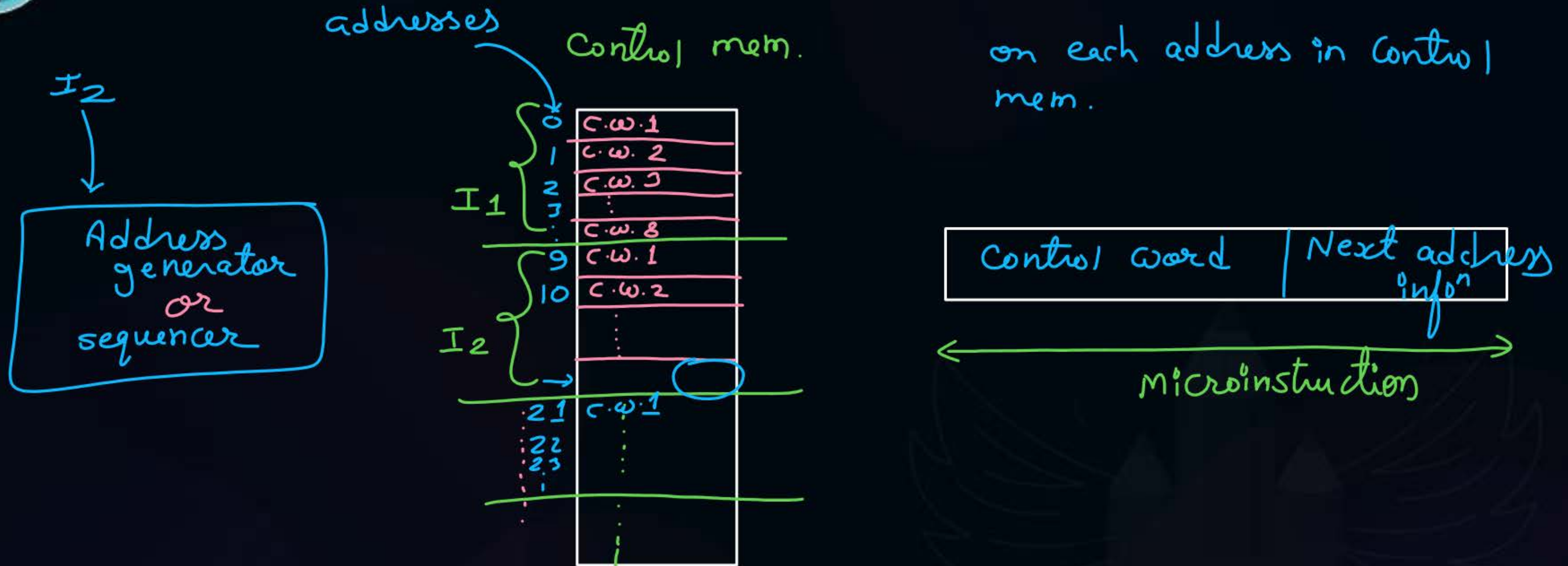
1. Updating the control logic is easy.
2. Easier to implement for complex computers also

Disadvantage: Slower than hardwired control unit.



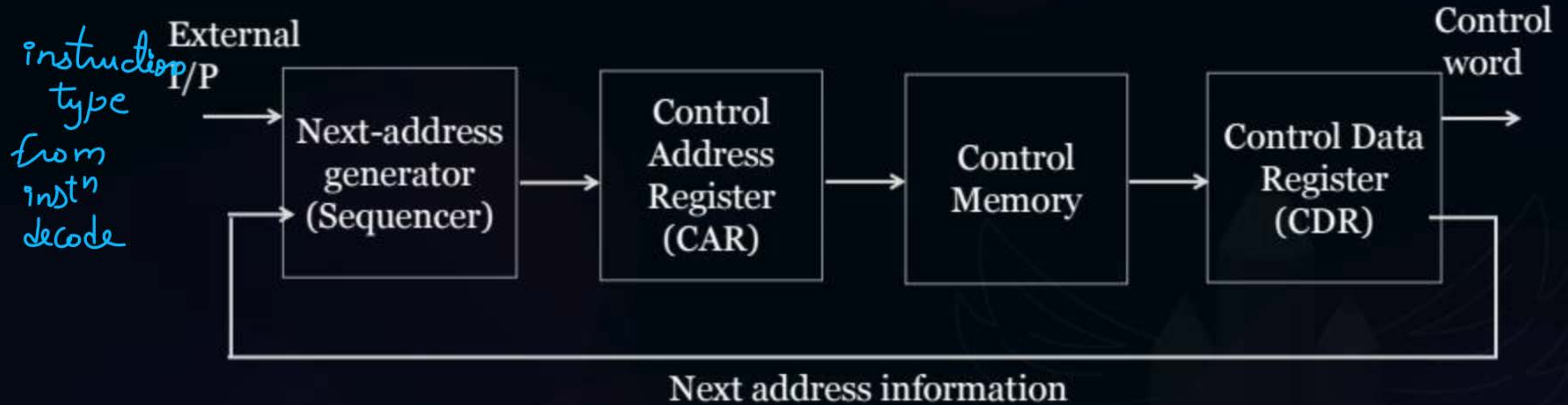


Topic : Micro-Programmed Control Unit





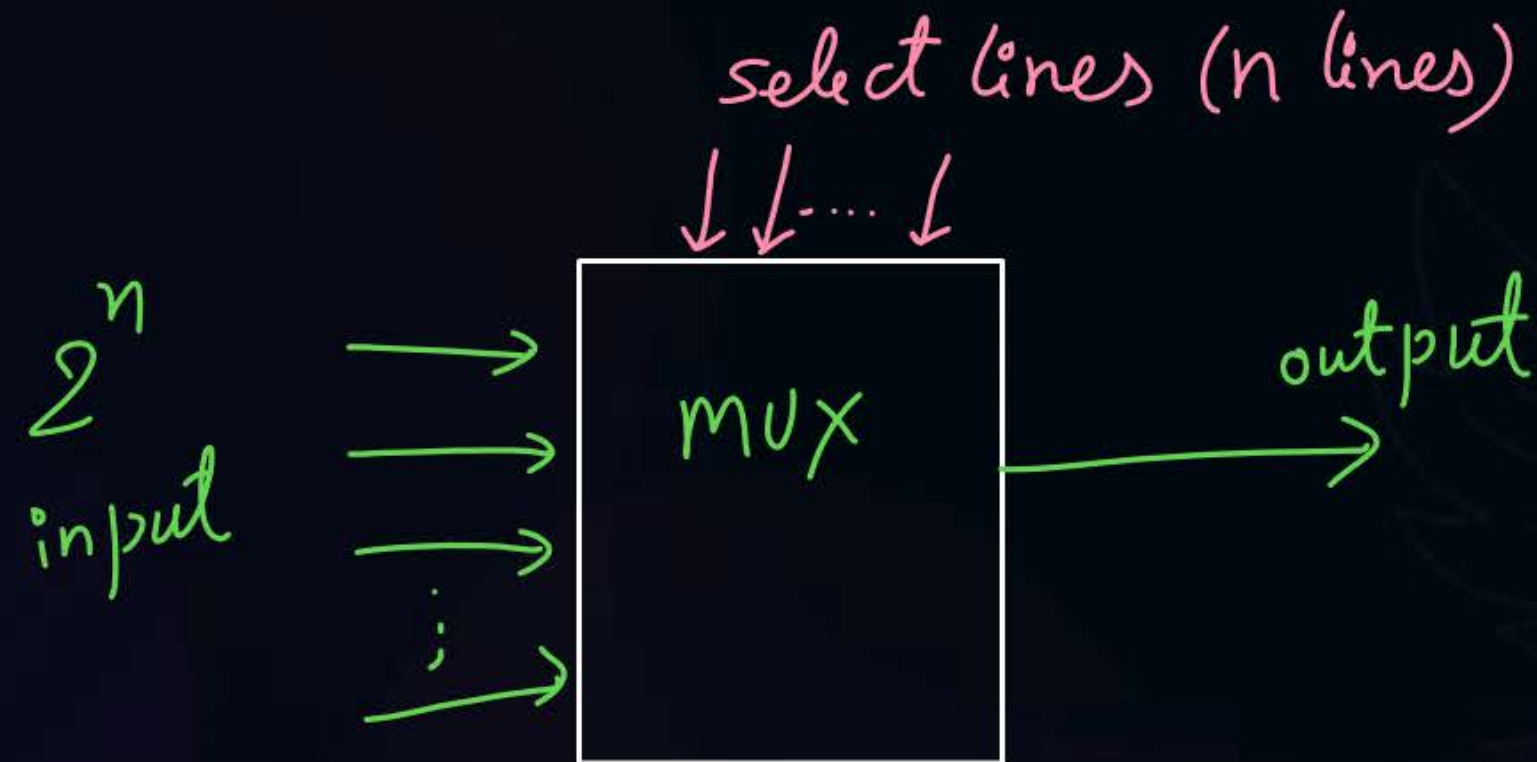
Topic : Control Word Sequencing





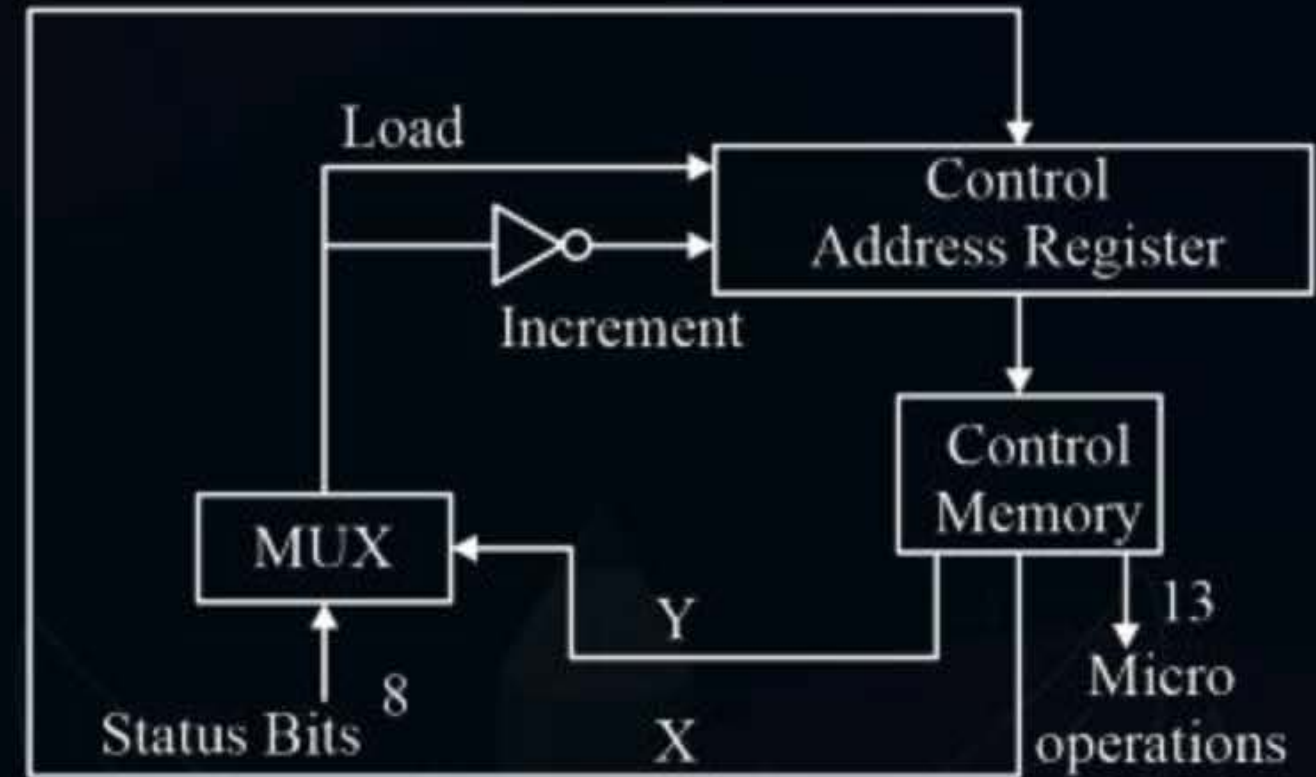
Topic : Control Word Sequencing

Standard format of micro-instⁿ



#Q. The microinstructions stored in the control memory of a processor have a width of 26 bits. Each microinstruction is divided into three fields: a micro-operation field of 13 bits, a next address field (X), and a MUX select field (Y). There are 8 status bits in the inputs of the MUX.

How many bits are there in the X and Y fields, and what is the size of the control memory in number of words? *no. of microinstns*

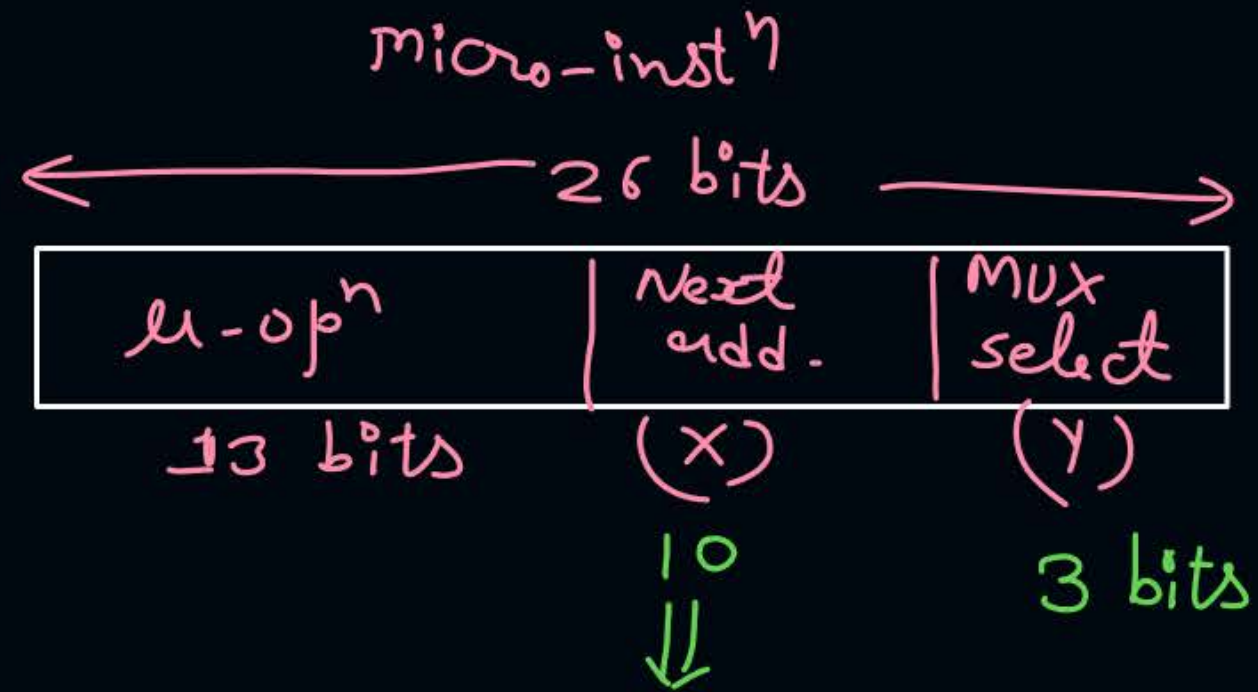


A ✓ 10, 3, 1024

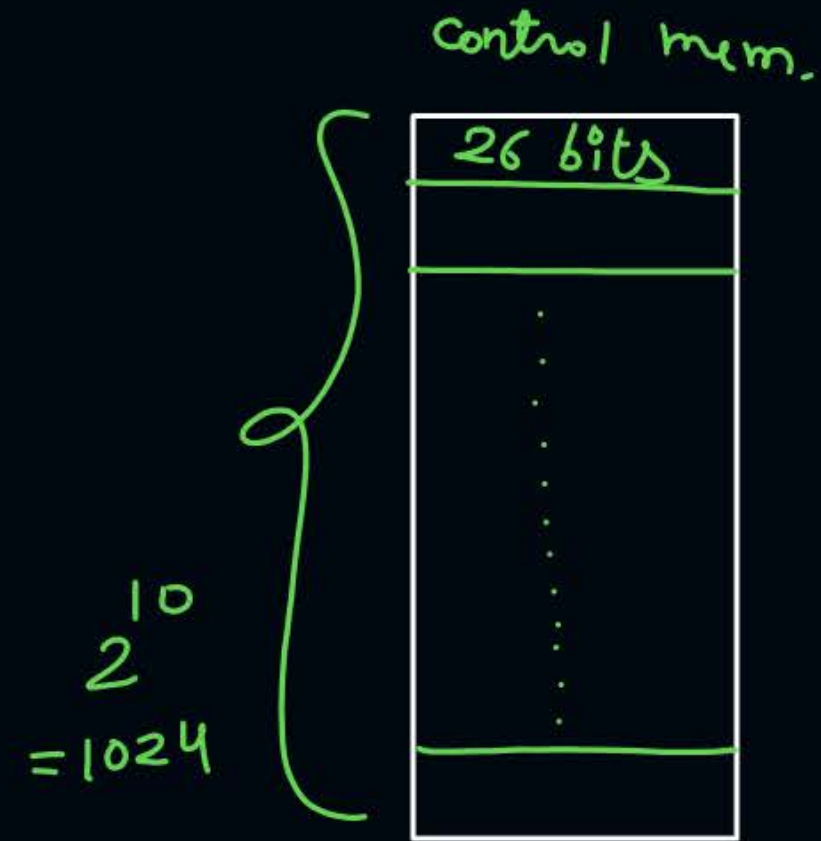
B 8, 5, 256

C 5, 8, 2048

D 10, 3, 512



no. of micro-inst^{ns}
in control mem. = 2^{10}



size of control mem.
= $2^{10} * 26 \text{ bits}$
= 26 k bits



Topic : Types of Microprogrammed Control Unit

Horizontal

- one bit for each control signal stored in control words in memory
- Larger size of control word

4 bits for 4 signals

S_0 active 0 0 0 1
 S_1 active 0 0 1 0
 S_2 — 0 1 0 0
 S_3 — 1 0 0 0

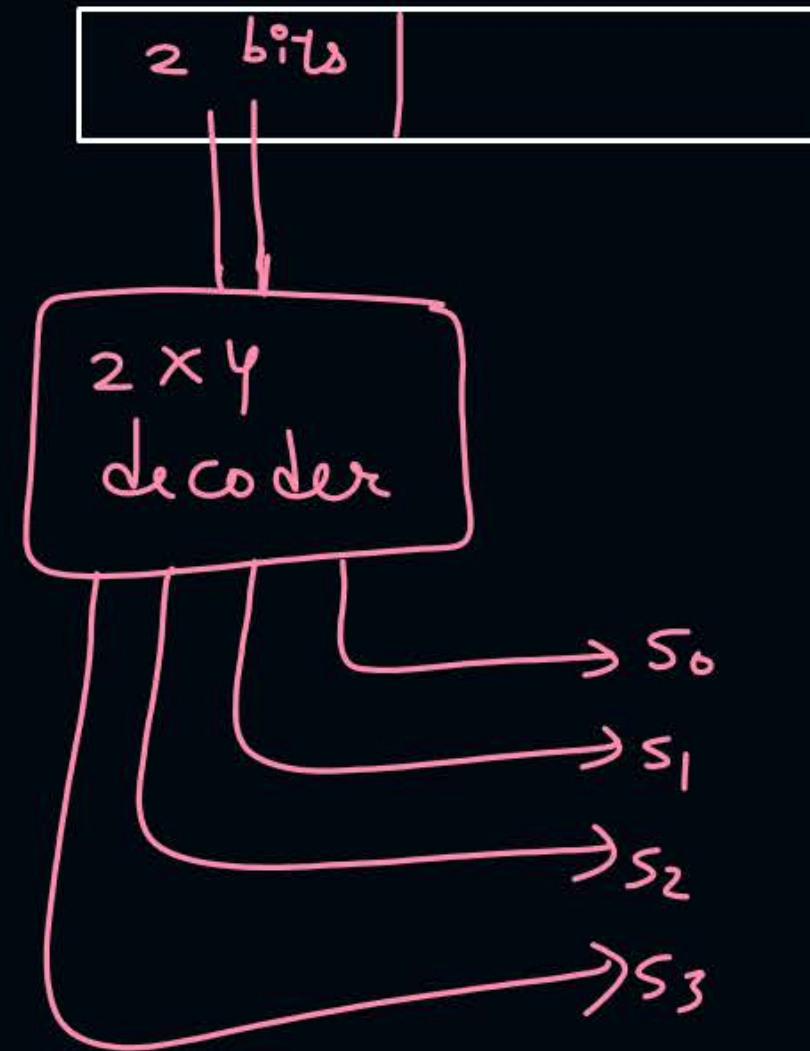
vertical

all control signals are divided into multiple groups, in such a way that from each group only one signal can be active at a time.

4 signals: S_3 S_2 S_1 S_0

Group 1 (2 bits)

0 0
0 1
1 0
1 1



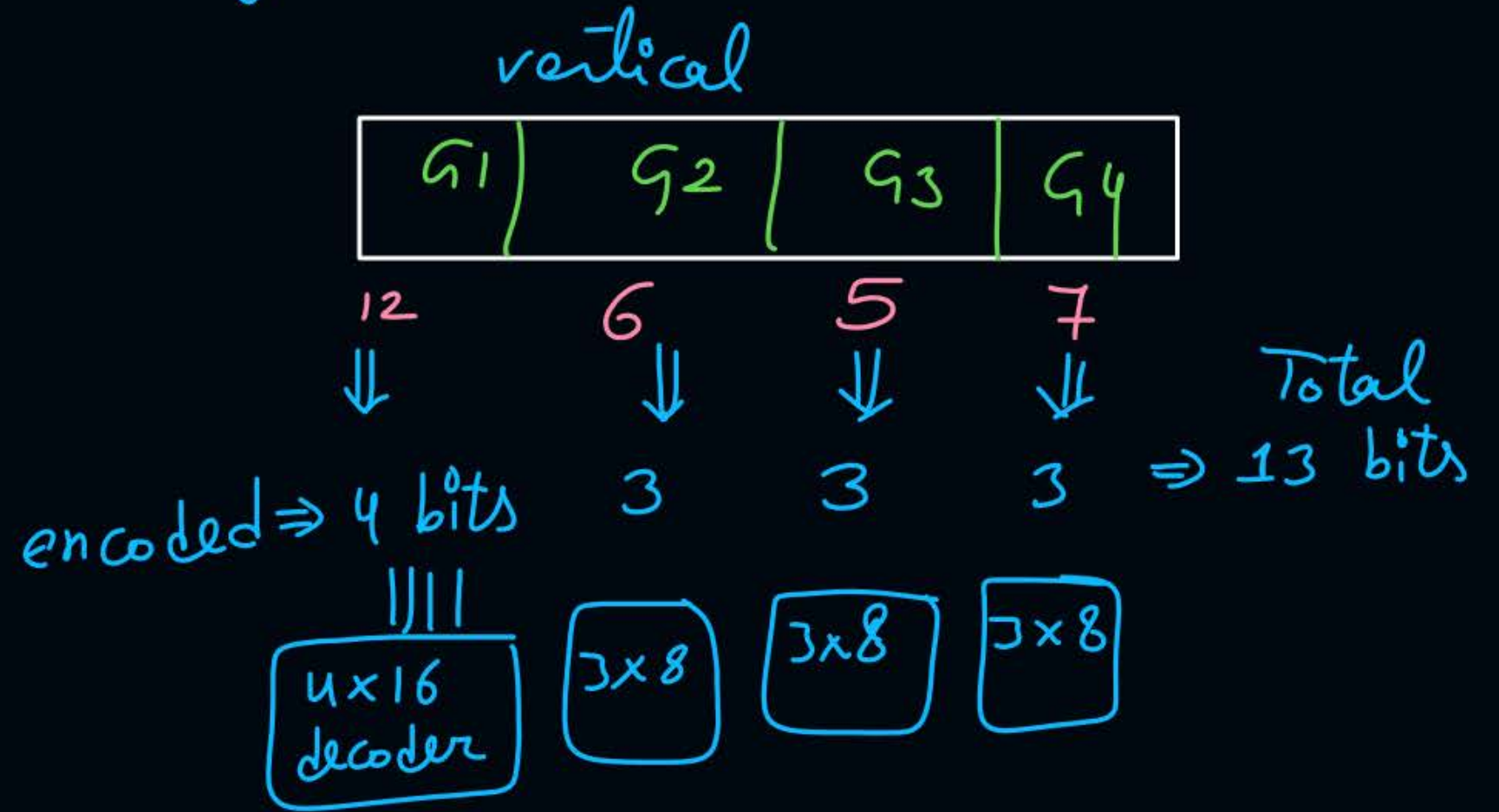
→ faster

- smaller size of control word
- decoders needed to generate signals
- slower

example:-

In a computer $\Rightarrow$ 30 control signals

each control word
in horizontal $\Rightarrow$ 30 bits





Topic : Speed Comparison



fastest

Hardwired

Horizontal
micro programmed

Slowest

vertical micro programmed



#Q. A control unit generates 120 control signals, which are divided into 6 groups of mutually exclusive signals as below:

Group1 = 30 $\Rightarrow$ 5 bits

Group2 = 13 $\Rightarrow$ 4 bits

Group3 = 12 $\Rightarrow$ 4 bits

Group4 = 3 $\Rightarrow$ 2 bits

Group5 = 27 $\Rightarrow$ 5 bits

Group6 = 35 $\Rightarrow$ 6 bits

26
bits

Horizontal = 120

vertical = 26

bits saved = 94 bits

How many bits can be saved by using vertical micro-programmed control unit as compared to horizontal one? for each control word?

ex:-

- A horizontal micro programmed CPU has to support = 16 inst^{ns}
- To execute each instⁿ, CPU has to generate 8 control words.

→ micro-instⁿ

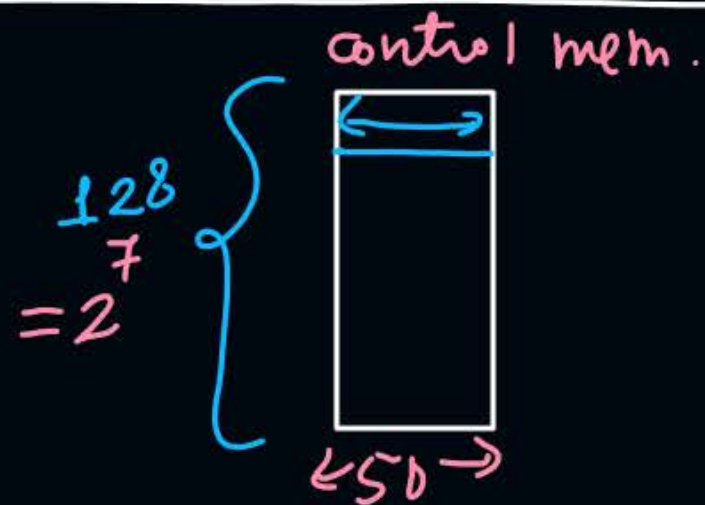
control word	MUX select	Address
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- There are 8 input in MUX
- There are 40 signals in CPU
- Size of control memory — bits?

Solⁿ

$$\begin{aligned}\text{Total no. of micro inst}^{\text{ns}} &= 16 * 8 \\ &= 128 = 2^7 \\ &\quad \downarrow \\ &\text{add.} = 7 \text{ bits}\end{aligned}$$

control word	MUX select	Add.
40	3	7
50 bits		



$$\begin{aligned}\text{Control mem. size} &= 128 * 50 \text{ bits} \\ &= 6400 \text{ bits}\end{aligned}$$

#Q. A micro-programmed control unit is required to support 64 instructions with each instruction need a sequence of 32 microoperations to execute completely. The control unit has microinstruction format with 3 fields: control signals, MUX select and next address field. The control unit has 134 signal to be stored in horizontal manner. The MUX has 8 inputs. The size of control memory to support such control unit is 296 k bits?

$$\text{no. of micro-inst}^{\text{ns}} = 64 * 32$$

$$= 2^{11}$$

$$\downarrow$$

$$\text{add.} = 11 \text{ bits}$$

$$\text{size of control unit} = 2^{11} * 148 \text{ bits}$$

$$= 296 \text{ k bits}$$

micro-instⁿ



#Q. A micro-programmed control unit is required to support 128 instructions with each instruction need a sequence of 64 microoperations to execute completely. The control unit has microinstruction format with 3 fields: control signals, MUX select and next address field. The control unit has 134 signal to be stored in horizontal manner. The MUX has 8 inputs. The size of control memory to support such control unit is _____ k bytes?

#Q. A micro-programmed control unit is required to generate a total of 25 control signals. Assume that during any microinstruction at most 2 control signals are active. Minimum number of bits required in the control word to generate the required control signals will be?





Topic : RISC vs CISC

S. No.	RISC (Reduced Instruction-Set Computer)	CISC (Complex Instruction-Set Computer)
1.	Less Number of Instructions Supported	More Number of Instructions
2.	Fixed Length Instructions	Variable Length Instructions
3.	Simple Instructions	Complex Instructions
4.	Simple and less number of addressing Modes	Complex and More number of addressing Modes
5.	Easy to implement using hardwired control unit	Difficult to implement using hardwired control unit



Topic : RISC vs CISC

S. No.	RISC (Reduced Instruction-Set Computer)	CISC (Complex Instruction-Set Computer)
6.	One Cycle per instruction	More than one cycle per instruction
7.	Register-to-Register arithmetic operation only $\Rightarrow$ Reg-based architecture	Register-to-Memory & Memory-to-Register arithmetic operations possible
8.	More Number of Registers (GPRs)	Less Number of Registers

#Q. Consider the following processor design characteristics.

I. Register-to register arithmetic operations only

II. Fixed-length instruction format

III. Hardwired control unit

Which of the characteristics above are used in the design of a RISC processor?

A

I and II only

B

II and III only

C

I and III only

D

✓ I, II and III



Topic : Byte Ordering

→ How multiple byte content is stored in memory.

2 bytes content



most significant byte

least significant byte

stored in mem. starting from add. 200

big endian

little endian

LS byte is stored on starting add.



ms byte is stored on starting add.

architectural decision to use one among two.



Topic : Byte Ordering

ex:-

200	10101100
201	00110010

if little endian

00110010 | 10101100

if big endian

10101100 | 00110010



2 mins Summary



Topic

Microprogrammed Control Unit

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RISC vs CISC

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Byte Ordering





Happy Learning

THANK - YOU /